

JESS Thermodynamic Database v8.9

Protactinium

24-Jan-23

Reaction No. 2754							
$\text{Pa}+4 + \text{H}+1_F-1 = \text{Pa}+4_F-1 + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO ₄	lgR:4.7(0.07SD)	5	CMN	437

Reaction No. 2755							
$\text{Pa}+4_F-1 + \text{H}+1_F-1 = \text{Pa}+4_F-1(2) + \text{H}+1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO ₄	lgR:3.5(0.07SD)	5	CMN	437

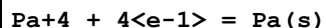
Reaction No. 4821							
$\text{PaO}2+1 + 4\langle\text{H}+1\rangle + 5\langle\text{e}-1\rangle = \text{Pa}(\text{s}) + 2\langle\text{H}2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-102.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.0(0.2SD)	0	MDG	775

Reaction No. 4822							
$\text{PaO}2(\text{s}) + 4\langle\text{H}+1\rangle + \text{e}-1 = \text{Pa}+3 + 2\langle\text{H}2\text{O}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.5(0.1SD)	0	MDG	775
3	25	0	Inf. Dilution	E:-2.0(2SF)	0	CRV	7761
4	25	0	Inf. Dilution	dH:138.3(4SF) kJ	2	CRV	7761

Reaction No. 4823							
$\text{Pa}+4 + \text{e}-1 = \text{Pa}+3$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-32.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.0(0.1SD)	0	MDG	775
3	25	0	Inf. Dilution	E:-1.9(2SF)	0	CRV	6330
4	25	0	Inf. Dilution	E:-1.9(2SF)	0	CRV	7761
5	25	0	Inf. Dilution	E:-1.4(2SF)	0	CRV	22551

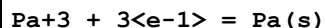
6	25	0	Inf. Dilution	dH:232.2 (4SF) kJ	2	CRV	7761
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Reaction No. 4824



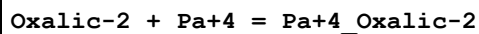
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:563.670 (23.174SD) kJ	1	CRV	32222
2	25	1	HCl	dG:135. (2.SD)	0	ENS	3006
3	25	0	Inf. Dilution	E:-1.7 (0.1SD)	0	MDG	775
4	25	0	Inf. Dilution	E:-1.49 (3SF)	2	CRV	6330
5	25	0	Inf. Dilution	E:-1.49 (3SF)	2	CRV	7761
6	25	0	Inf. Dilution	E:-1.46 (3SF)	0	CRV	22551
7	25	1	HCl	E:-1.5 (0.06SD)	2	ENS	3006
8	25	0	Inf. Dilution	dH:636.0 (4SF) kJ	0	CRV	7761
9	25	0	Inf. Dilution	dH:620.000 (13.000SD) kJ	1	CRV	32222
10	25	1	HCl	dH:148. (2.SD)	1	ENS	3006

Reaction No. 4825



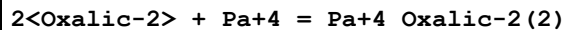
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.0 (0.05SD)	0	MDG	775
2	25	0	Inf. Dilution	E:-1.34 (3SF)	5	CRV	6330
3	25	0	Inf. Dilution	E:-1.49 (3SF)	0	ENS	7276
4	25	0	Inf. Dilution	E:-1.34 (3SF)	3	CRV	7761
5	25	0	Inf. Dilution	dH:400.0 (4SF) kJ	3	CRV	7761

Reaction No. 40394



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.7 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:10.7 (3SF)	5	EOD	931

Reaction No. 40395



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.3 (4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:20.3 (3SF)	5	EOD	931

Reaction No. 40396							
3<Oxalic-2> + Pa+4 = Pa+4_Oxalic-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.5(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:26.5(3SF)	5	EOD	931

Reaction No. 40397							
4<Oxalic-2> + Pa+4 = Pa+4_Oxalic-2(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.2(4SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:29.2(3SF)	5	EOD	931

Reaction No. 40913							
PaO2+1_OH-1(3) + H+1_234TriOHPentDioic-2 = H+1_PaO2+1_OH-1(3)_234TriOHPentDioic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.25	Unknown	lgK:2.80(3SF)	3	MIX	931

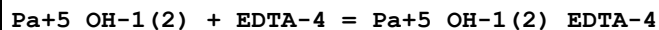
Reaction No. 42055							
Oxalic-2 + Pa+5 = Pa+5_Oxalic-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HNO3	lgK:1.81(3SF)	5	MIX	4421
2	23	2	HNO3	lgK:1.813(4SF)	5	MIX	4421
3	23	3	HNO3	lgK:1.623(4SF)	5	MIX	4421
4	23	4	HNO3	lgK:1.556(4SF)	5	MIX	4421
5	23	6	HNO3	lgK:1.380(4SF)	5	MIX	4421
6	25	0	Inf. Dilution	lgK:1.8(2SF)	1	ESC	

Reaction No. 42056							
2<Oxalic-2> + Pa+5 = Pa+5_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HNO3	lgK:2.505(4SF)	0	MIX	4421
2	23	2	HNO3	lgK:2.748(4SF)	0	MIX	4421
3	23	3	HNO3	lgK:2.079(4SF)	0	MIX	4421
4	25	0	Inf. Dilution	lgK:2.5(2SF)	1	EES	

Reaction No. 42057							
3<Oxalic-2> + Pa+5 = Pa+5_Oxalic-2(3)							

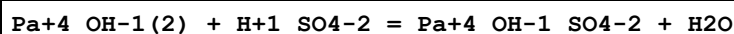
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	HNO3	lgK:4.338 (4SF)	5	MIX	4421
2	25	0	Inf. Dilution	lgK:4.3 (2SF)	1	ESC	

Reaction No. 44867



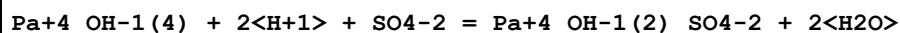
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	Unknown	lgK:22.1 (0.1SD)	4	MDS	906
2	25	0	Inf. Dilution	lgK:22.1 (3SF)	1	ESC	

Reaction No. 53315



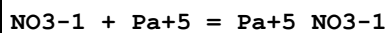
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10	0.5	Unknown	lgK:2.51 (3SF)	5	MDS	931
2	10	0.5	Unknown	lgK:2.50 (3SF)	5	MDS	5398

Reaction No. 53316



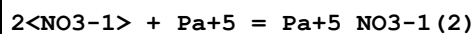
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	17	1.38	NaClO4	lgK:0.81 (2SF)	4	MDS	5398

Reaction No. 53840



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:-0.17 (2SF)	5	MCE	931
2	20	2	HClO4	lgK:-0.10 (2SF)	5	MCE	931
3	20	4	HClO4	lgK:-0.20 (2SF)	5	MCE	931
4	25	1	HClO4	lgK:0.16 (2SF)	5	MDS	931

Reaction No. 53841



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:0.48 (2SF)	5	MCE	931
2	20	2	HClO4	lgK:-0.13 (2SF)	5	MCE	931
3	20	4	HClO4	lgK:-0.68 (2SF)	5	MCE	931
4	25	0	Inf. Dilution	lgK:0.5 (1SF)	1	ESC	

Reaction No. 53842							
$4\text{<NO3-1>} + \text{Pa+5} = \text{Pa+5_NO3-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	1	HClO4	lgK:1.08 (3SF)	5	MCE	931

Reaction No. 53843							
$\text{Pa+5_NO3-1} + \text{NO3-1} = \text{Pa+5_NO3-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.1 (2SF)	1	ELC	
2	25	1	HClO4	lgK:-1.15 (3SF)	3	MDS	931

Reaction No. 54060							
$\text{Pa+4} + \text{H+1_SO4-2} = \text{Pa+4_SO4-2} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:1.62 (3SF)	5	MDS	931

Reaction No. 54061							
$\text{Pa+4} + 2\text{<H+1_SO4-2>} = \text{Pa+4_SO4-2(2)} + 2\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:2.18 (3SF)	5	MDS	931

Reaction No. 54062							
$\text{Pa+5} + \text{H+1_SO4-2} = \text{Pa+5_SO4-2} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:2.08 (3SF)	5	MDS	931

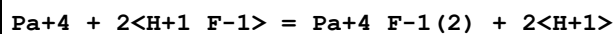
Reaction No. 54063							
$\text{Pa+5_SO4-2} + \text{H+1_SO4-2} = \text{Pa+5_SO4-2(2)} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:0.23 (2SF)	5	MDS	931

Reaction No. 54064							
$\text{PaO+3_OH-1} + \text{H+1_SO4-2} = \text{PaO+3_SO4-2} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgK:1.29 (3SF)	4	MDS	931

Reaction No. 54065							
$\text{PaO+3_OH-1} + 2\text{<H+1_SO4-2>} = \text{PaO+3_SO4-2(2)} + \text{H+1} + \text{H2O}$							

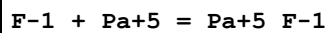
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:2.51 (3SF)	4	MDS	931

Reaction No. 54124



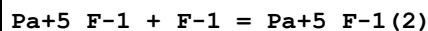
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgR:8.26 (3SF)	4	MDS	931

Reaction No. 54125



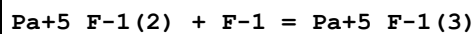
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.4 (2SF)	5	MQH	931

Reaction No. 54126



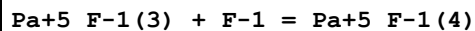
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:5.0 (2SF)	5	MQH	931

Reaction No. 54127



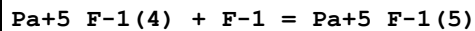
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:4.9 (2SF)	5	MQH	931

Reaction No. 54128



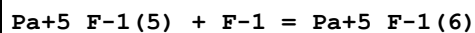
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:4.8 (2SF)	5	MQH	931

Reaction No. 54129



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:4.5 (2SF)	5	MQH	931

Reaction No. 54130



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:4.4 (2SF)	5	MQH	931

Reaction No. 54131							
$\text{Pa+5_F-1(6)} + \text{F-1} = \text{Pa+5_F-1(7)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:3.7(2SF)	5	MQH	931

Reaction No. 54132							
$\text{Pa+5_F-1(7)} + \text{F-1} = \text{Pa+5_F-1(8)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaClO4	lgK:1.7(2SF)	5	MQH	931

Reaction No. 54133							
$\text{PaO+3_OH-1} + \text{H+1_F-1} = \text{PaO+3_F-1} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	NaClO4	lgK:3.56(3SF)	5	MDS	931

Reaction No. 54134							
$\text{Pa+5} + \text{H+1_F-1} = \text{Pa+5_F-1} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:3.95(3SF)	4	MDS	931

Reaction No. 54135							
$\text{Pa+5_F-1} + \text{H+1_F-1} = \text{Pa+5_F-1(2)} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:3.48(3SF)	4	MDS	931

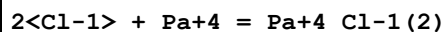
Reaction No. 54136							
$\text{Pa+5_F-1(2)} + \text{H+1_F-1} = \text{Pa+5_F-1(3)} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:3.04(3SF)	4	MDS	931

Reaction No. 54179							
$\text{Pa+4} + \text{H+1_Cl-1} = \text{Pa+4_Cl-1} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	3	HClO4	lgR:0.0(1SF)	3	MDS	931

Reaction No. 54180							
$\text{Pa+4} + 2\langle\text{H+1_Cl-1}\rangle = \text{Pa+4_Cl-1(2)} + 2\langle\text{H+1}\rangle$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

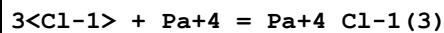
1	23	3	HClO4	lgR:0.0 (1SF)	3	MDS	931
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Reaction No. 54181



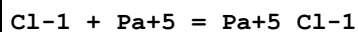
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.19 (3SF)	4	EBU	6372
2	25	3	HClO4	lgK:0.0 (1SF)	4	MDS	931

Reaction No. 54182



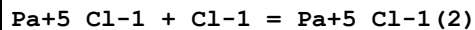
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.94 (3SF)	4	EBU	6372
2	25	3	HClO4	lgK:0.0 (1SF)	3	MDS	931

Reaction No. 54183



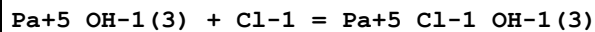
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:0.21 (2SF)	4	MDS	931

Reaction No. 54184



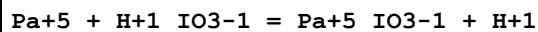
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgK:-0.89 (2SF)	4	MDS	931

Reaction No. 54185



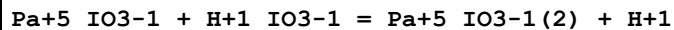
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	HClO4	lgK:0.08 (1SF)	4	MDS	931

Reaction No. 54385



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:2.11 (3SF)	5	MDS	931

Reaction No. 54386



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	HClO4	lgR:1.54 (3SF)	5	MDS	931

Reaction No. 55649							
Pa+3 + e-1 = Pa+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Unknown	E:-5. (0.5SD)	4	ENS	5256

Reaction No. 56065							
Pa+4 + H2O = Pa+4_OH-1 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.41 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:0.84 (2SF)	4	CRV	6478
3	25	0	Inf. Dilution	lgK:0.84 (0.10SD)	4	CRV	32222
4	25	0	CHEMVAL4	lgK:0.66 (0.01SD)	4	ENS	5156
5	25	0	Inf. Dilution	dH:8.7 (2SF)	3	CRV	6478

Reaction No. 56066							
Pa+4 + 2<H2O> = Pa+4_OH-1 (2) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.0 (1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.37 (3SF)	0	EBU	6372
3	25	0	Inf. Dilution	lgK:-0.02 (0.20SD)	4	CRV	32222
4	25	0	CHEMVAL4	lgK:0.80 (0.01SD)	0	ENS	5156

Reaction No. 56067							
Pa+4 + 3<H2O> = Pa+4_OH-1 (3) + 3<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.75 (3SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:-1.50 (0.30SD)	4	CRV	32222
3	25	0	CHEMVAL4	lgK:-1.50 (0.01SD)	2	ENS	5156

Reaction No. 56310							
PaO2(s) + 4<H+1> = Pa+4 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.7 (1SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:0.6 (3.SD)	2	CRV	6478
3	25	0	Inf. Dilution	lgK:0.60 (2SF)	4	CRV	32222
4	25	0	CHEMVAL4	lgK:0.60 (0.01SD)	2	ENS	5156

Reaction No. 56311							
$\text{Pa+4}_{\text{Cl-1}(4)}(\text{s}) = \text{Pa+4} + 4\text{Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL4	$\lg K: 24.10 (0.01\text{SD})$	4	ENS	5156

Reaction No. 56341							
$\text{Pa}(\text{s}) + \text{O}_2(\text{g}) = \text{PaO}_2(\text{s})$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 183.0 (4\text{SF})$	1	ELC	
2	25	0	Inf. Dilution	$dG: -252. (2.\text{SD})$	0	ENS	3006
3	25	0	Inf. Dilution	$dH: -265. (2.\text{SD})$	2	ENS	3006

Reaction No. 58712							
$\text{NTA-3} + \text{PaO+3} = \text{PaO+3}_{\text{NTA-3}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO4	$\lg K: 15. (2\text{SF})$	2	MDS	30786
Note (1): $\lg K$ varies with acidity							

Reaction No. 60444							
$\text{Pa+4} + \text{OH-1} = \text{Pa+4}_{\text{OH-1}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 14.8 (3\text{SF})$	4	CRV	2556
2	25	3	LiClO4	$\lg K: 14.04 (4\text{SF})$	5	CRV	2556

Reaction No. 60445							
$\text{Pa+4} + 2\text{OH-1} = \text{Pa+4}_{\text{OH-1}(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 28.0 (3\text{SF})$	4	CRV	2556
2	25	3	LiClO4	$\lg K: 27.8 (3\text{SF})$	5	CRV	2556

Reaction No. 60446							
$\text{Pa+4} + 3\text{OH-1} = \text{Pa+4}_{\text{OH-1}(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	$\lg K: 40.7 (3\text{SF})$	2	EOR	
2	25	3	LiClO4	$\lg K: 40.7 (3\text{SF})$	5	CRV	2556

Reaction No. 60447							
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Pa+4 + 4<OH-1> = Pa+4_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	LiClO4	lgK:51.4 (3SF)	5	CRV	2556

Reaction No. 60453							
Pa+5_OH-1(3) + OH-1 = Pa+5_OH-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.5 (3SF)	4	CRV	2556
2	25	3	HClO4	lgK:13.13 (4SF)	5	CRV	2556

Reaction No. 60454							
Pa+5_OH-1(4) + OH-1 = Pa+5_OH-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.5 (2SF)	4	CRV	2556
2	25	3	HClO4	lgK:9.68 (3SF)	5	CRV	2556

Reaction No. 63984							
Pa+4 + 4<H2O> = Pa+4_OH-1(4) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.0 (2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-12.51 (4SF)	0	EBU	6372

Reaction No. 63985							
Pa+4 + 5<H2O> = Pa+4_OH-1(5) + 5<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.61 (4SF)	1	EBU	6372

Reaction No. 63986							
Pa+4 + 6<H2O> = Pa+4_OH-1(6) + 6<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.04 (4SF)	1	EBU	6372

Reaction No. 63987							
4<Pa+4> + 12<H2O> = Pa+4(4)_OH-1(12) + 12<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-30.89 (4SF)	1	EBU	6372

Reaction No. 63988							
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6<Pa+4> + 15<H2O> = Pa+4(6)_OH-1(15) + 15<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.66(4SF)	1	EBU	6372

Reaction No. 63989							
F-1 + Pa+4 = Pa+4_F-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.13(3SF)	1	EBU	6372

Reaction No. 63990							
2<F-1> + Pa+4 = Pa+4_F-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.01(4SF)	1	EBU	6372

Reaction No. 63991							
3<F-1> + Pa+4 = Pa+4_F-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.77(4SF)	1	EBU	6372

Reaction No. 63992							
4<F-1> + Pa+4 = Pa+4_F-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.44(4SF)	1	EBU	6372

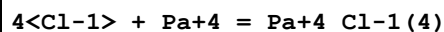
Reaction No. 63993							
5<F-1> + Pa+4 = Pa+4_F-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.06(4SF)	1	EBU	6372

Reaction No. 63994							
6<F-1> + Pa+4 = Pa+4_F-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.66(4SF)	1	EBU	6372

Reaction No. 63995							
Cl-1 + Pa+4 = Pa+4_Cl-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.25(3SF)	1	EBU	6372

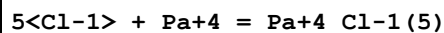
2	25	3	HClO4	lgK:0.0(1SF)	6	CRV	552
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Reaction No. 63996



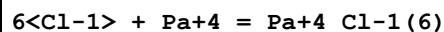
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.55(3SF)	1	EBU	6372

Reaction No. 63997



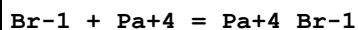
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.06(3SF)	1	EBU	6372

Reaction No. 63998



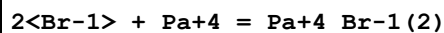
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.48(3SF)	1	EBU	6372

Reaction No. 63999



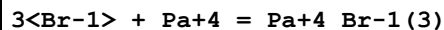
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.76(2SF)	1	EBU	6372

Reaction No. 64000



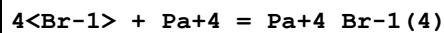
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.21(3SF)	1	EBU	6372

Reaction No. 64001



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.49(3SF)	1	EBU	6372

Reaction No. 64002



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.64(3SF)	1	EBU	6372

Reaction No. 64003

5<Br-1> + Pa+4 = Pa+4_Br-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.70(3SF)	1	EBU	6372

Reaction No. 64004							
6<Br-1> + Pa+4 = Pa+4_Br-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.67(3SF)	1	EBU	6372

Reaction No. 64005							
I-1 + Pa+4 = Pa+4_I-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.46(2SF)	1	EBU	6372

Reaction No. 64006							
2<I-1> + Pa+4 = Pa+4_I-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.63(2SF)	1	EBU	6372

Reaction No. 64007							
3<I-1> + Pa+4 = Pa+4_I-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.61(2SF)	1	EBU	6372

Reaction No. 64008							
4<I-1> + Pa+4 = Pa+4_I-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.48(2SF)	1	EBU	6372

Reaction No. 64009							
5<I-1> + Pa+4 = Pa+4_I-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.24(2SF)	1	EBU	6372

Reaction No. 64010							
6<I-1> + Pa+4 = Pa+4_I-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.07(1SF)	1	EBU	6372

Reaction No. 64011							
IO3-1 + Pa+4 = Pa+4_IO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.14 (3SF)	1	EBU	6372

Reaction No. 64012							
2<IO3-1> + Pa+4 = Pa+4_IO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.62 (3SF)	1	EBU	6372

Reaction No. 64013							
3<IO3-1> + Pa+4 = Pa+4_IO3-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.70 (3SF)	1	EBU	6372

Reaction No. 64014							
4<IO3-1> + Pa+4 = Pa+4_IO3-1 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.47 (3SF)	1	EBU	6372

Reaction No. 64015							
5<IO3-1> + Pa+4 = Pa+4_IO3-1 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.00 (4SF)	1	EBU	6372

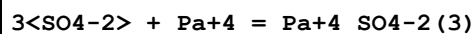
Reaction No. 64016							
6<IO3-1> + Pa+4 = Pa+4_IO3-1 (6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.31 (4SF)	1	EBU	6372

Reaction No. 64017							
SO4-2 + Pa+4 = Pa+4_SO4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.87 (3SF)	1	EBU	6372

Reaction No. 64018							
2<SO4-2> + Pa+4 = Pa+4_SO4-2 (2)							

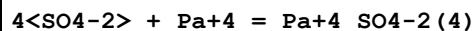
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.35 (3SF)	1	EBU	6372

Reaction No. 64019



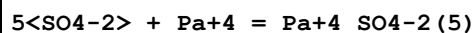
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.68 (4SF)	1	EBU	6372

Reaction No. 64020



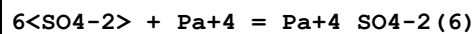
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.96 (4SF)	1	EBU	6372

Reaction No. 64021



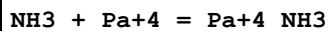
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.26 (4SF)	1	EBU	6372

Reaction No. 64022



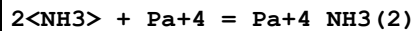
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.60 (4SF)	1	EBU	6372

Reaction No. 64023



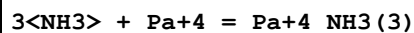
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.27 (3SF)	1	EBU	6372

Reaction No. 64024



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.08 (4SF)	1	EBU	6372

Reaction No. 64025



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.55 (4SF)	1	EBU	6372

Reaction No. 64026							
4<NH3> + Pa+4 = Pa+4_NH3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.73(4SF)	1	EBU	6372

Reaction No. 64027							
5<NH3> + Pa+4 = Pa+4_NH3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.65(4SF)	1	EBU	6372

Reaction No. 64028							
6<NH3> + Pa+4 = Pa+4_NH3(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.34(4SF)	1	EBU	6372

Reaction No. 64029							
NO3-1 + Pa+4 = Pa+4_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.79(3SF)	1	EBU	6372

Reaction No. 64030							
2<NO3-1> + Pa+4 = Pa+4_NO3-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.85(3SF)	1	EBU	6372

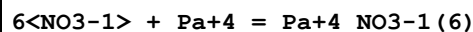
Reaction No. 64031							
3<NO3-1> + Pa+4 = Pa+4_NO3-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.42(3SF)	1	EBU	6372

Reaction No. 64032							
4<NO3-1> + Pa+4 = Pa+4_NO3-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.61(3SF)	1	EBU	6372

Reaction No. 64033							
5<NO3-1> + Pa+4 = Pa+4_NO3-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

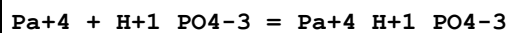
1	25	0	Inf. Dilution	lgK:3.47 (3SF)	1	EBU	6372
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Reaction No. 64034



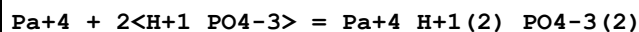
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.04 (3SF)	1	EBU	6372

Reaction No. 64035



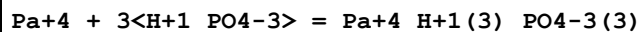
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.64 (4SF)	1	EBU	6372

Reaction No. 64036



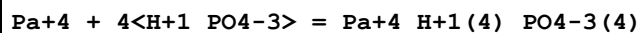
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.68 (4SF)	1	EBU	6372

Reaction No. 64037



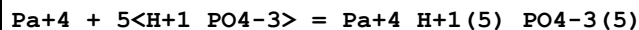
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.38 (4SF)	1	EBU	6372

Reaction No. 64038



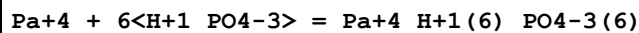
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1	25	0	Inf. Dilution	lgK:41.82 (4SF)	1	EBU	6372

Reaction No. 64039



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:49.07 (4SF)	1	EBU	6372

Reaction No. 64040



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.16 (4SF)	1	EBU	6372

Reaction No. 64041

Pa+4 + H+1(2)_PO4-3 = Pa+4_H+1(2)_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.65(3SF)	1	EBU	6372

Reaction No. 64042							
Pa+4 + 2<H+1(2)_PO4-3> = Pa+4_H+1(4)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.71(3SF)	1	EBU	6372

Reaction No. 64043							
Pa+4 + 3<H+1(2)_PO4-3> = Pa+4_H+1(6)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.41(3SF)	1	EBU	6372

Reaction No. 64044							
Pa+4 + 4<H+1(2)_PO4-3> = Pa+4_H+1(8)_PO4-3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.87(3SF)	1	EBU	6372

Reaction No. 64045							
Pa+4 + 5<H+1(2)_PO4-3> = Pa+4_H+1(10)_PO4-3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.14(3SF)	1	EBU	6372

Reaction No. 64046							
Pa+4 + 6<H+1(2)_PO4-3> = Pa+4_H+1(12)_PO4-3(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.26(3SF)	1	EBU	6372

Reaction No. 64047							
CO3-2 + Pa+4 = Pa+4_CO3-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.53(4SF)	1	EBU	6372

Reaction No. 64048							
2<CO3-2> + Pa+4 = Pa+4_CO3-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.12(4SF)	1	EBU	6372

Reaction No. 64049							
$3\text{<CO3-2>} + \text{Pa+4} = \text{Pa+4_CO3-2(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.03(4SF)	1	EBU	6372

Reaction No. 64050							
$4\text{<CO3-2>} + \text{Pa+4} = \text{Pa+4_CO3-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.34(4SF)	1	EBU	6372

Reaction No. 64051							
$5\text{<CO3-2>} + \text{Pa+4} = \text{Pa+4_CO3-2(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.12(4SF)	1	EBU	6372

Reaction No. 64052							
$6\text{<CO3-2>} + \text{Pa+4} = \text{Pa+4_CO3-2(6)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:58.40(4SF)	1	EBU	6372

Reaction No. 64053							
$\text{Pa+4} + \text{H+1_CO3-2} = \text{Pa+4_H+1_CO3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.72(3SF)	1	EBU	6372

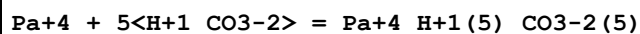
Reaction No. 64054							
$\text{Pa+4} + 2\text{<H+1_CO3-2>} = \text{Pa+4_H+1(2)_CO3-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.31(4SF)	1	EBU	6372

Reaction No. 64055							
$\text{Pa+4} + 3\text{<H+1_CO3-2>} = \text{Pa+4_H+1(3)_CO3-2(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.03(4SF)	1	EBU	6372

Reaction No. 64056							
$\text{Pa+4} + 4\text{<H+1_CO3-2>} = \text{Pa+4_H+1(4)_CO3-2(4)}$							

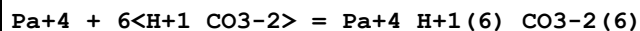
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.98 (4SF)	1	EBU	6372

Reaction No. 64057



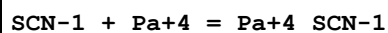
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.21 (4SF)	1	EBU	6372

Reaction No. 64058



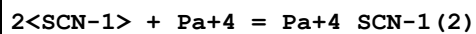
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.76 (4SF)	1	EBU	6372

Reaction No. 64059



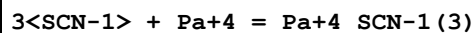
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.33 (3SF)	1	EBU	6372

Reaction No. 64060



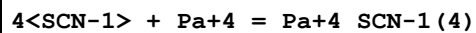
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.24 (3SF)	1	EBU	6372

Reaction No. 64061



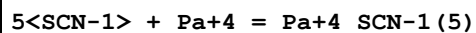
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.85 (3SF)	1	EBU	6372

Reaction No. 64062



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.21 (3SF)	1	EBU	6372

Reaction No. 64063



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.36 (3SF)	1	EBU	6372

Reaction No. 64064							
6<SCN-1> + Pa+4 = Pa+4_SCN-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.31(3SF)	1	EBU	6372

Reaction No. 64065							
Acetic-1 + Pa+4 = Pa+4_Acetic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.69(3SF)	1	EBU	6372

Reaction No. 64066							
2<Acetic-1> + Pa+4 = Pa+4_Acetic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.33(4SF)	1	EBU	6372

Reaction No. 64067							
3<Acetic-1> + Pa+4 = Pa+4_Acetic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.18(4SF)	1	EBU	6372

Reaction No. 64068							
4<Acetic-1> + Pa+4 = Pa+4_Acetic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.34(4SF)	1	EBU	6372

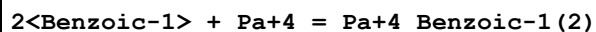
Reaction No. 64069							
5<Acetic-1> + Pa+4 = Pa+4_Acetic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.86(4SF)	1	EBU	6372

Reaction No. 64070							
6<Acetic-1> + Pa+4 = Pa+4_Acetic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.78(4SF)	1	EBU	6372

Reaction No. 64071							
Benzoic-1 + Pa+4 = Pa+4_Benzoic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

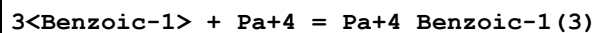
1	25	0	Inf. Dilution	lgK:6.35 (3SF)	1	EBU	6372
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Reaction No. 64072



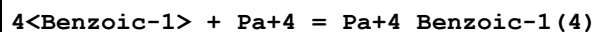
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.79 (4SF)	1	EBU	6372

Reaction No. 64073



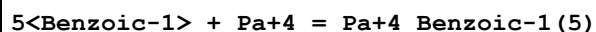
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.59 (4SF)	1	EBU	6372

Reaction No. 64074



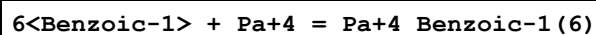
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.83 (4SF)	1	EBU	6372

Reaction No. 64075



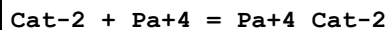
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.59 (4SF)	1	EBU	6372

Reaction No. 64076



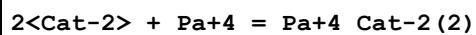
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.89 (4SF)	1	EBU	6372

Reaction No. 64077



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.75 (4SF)	1	EBU	6372

Reaction No. 64078



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.27 (4SF)	1	EBU	6372

Reaction No. 64079

3<Cat-2> + Pa+4 = Pa+4_Cat-2(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:71.22(4SF)	1	EBU	6372

Reaction No. 64080							
EtDiAm + Pa+4 = Pa+4_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.01(4SF)	1	EBU	6372

Reaction No. 64081							
2<EtDiAm> + Pa+4 = Pa+4_EtDiAm(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.87(4SF)	1	EBU	6372

Reaction No. 64082							
3<EtDiAm> + Pa+4 = Pa+4_EtDiAm(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.28(4SF)	1	EBU	6372

Reaction No. 64083							
Gly-1 + Pa+4 = Pa+4_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.42(4SF)	1	EBU	6372

Reaction No. 64084							
2<Gly-1> + Pa+4 = Pa+4_Gly-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.39(4SF)	1	EBU	6372

Reaction No. 64085							
3<Gly-1> + Pa+4 = Pa+4_Gly-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.12(4SF)	1	EBU	6372

Reaction No. 64086							
Glycolic-1 + Pa+4 = Pa+4_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.09(3SF)	1	EBU	6372

Reaction No. 64087							
2<Glycolic-1> + Pa+4 = Pa+4_Glycolic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.08(3SF)	1	EBU	6372

Reaction No. 64088							
3<Glycolic-1> + Pa+4 = Pa+4_Glycolic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.19(4SF)	1	EBU	6372

Reaction No. 64089							
4<Glycolic-1> + Pa+4 = Pa+4_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.55(4SF)	1	EBU	6372

Reaction No. 64090							
5<Glycolic-1> + Pa+4 = Pa+4_Glycolic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.19(4SF)	1	EBU	6372

Reaction No. 64091							
6<Glycolic-1> + Pa+4 = Pa+4_Glycolic-1(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.17(4SF)	1	EBU	6372

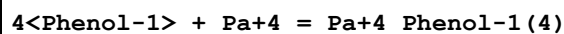
Reaction No. 64092							
Phenol-1 + Pa+4 = Pa+4_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.02(3SF)	1	EBU	6372

Reaction No. 64093							
2<Phenol-1> + Pa+4 = Pa+4_Phenol-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.36(4SF)	1	EBU	6372

Reaction No. 64094							
3<Phenol-1> + Pa+4 = Pa+4_Phenol-1(3)							

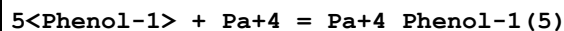
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.14 (4SF)	1	EBU	6372

Reaction No. 64095



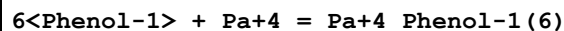
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.41 (4SF)	1	EBU	6372

Reaction No. 64096



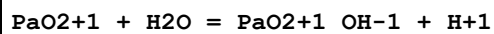
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.20 (4SF)	1	EBU	6372

Reaction No. 64097



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:45.53 (4SF)	1	EBU	6372

Reaction No. 64098



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.82 (4SF)	0	EBU	6372
2	25	0	Inf. Dilution	lgK:-4.50 (3SF)	4	CRV	32222

Reaction No. 64099



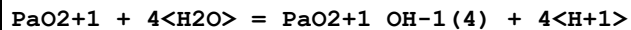
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-22.42 (4SF)	1	EBU	6372

Reaction No. 64100



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-34.68 (4SF)	1	EBU	6372

Reaction No. 64101



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-47.55 (4SF)	1	EBU	6372

Reaction No. 64102							
$\text{PaO2+1} + 5\text{<H2O>} = \text{PaO2+1_OH-1(5)} + 5\text{<H+1>}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-61.01(4SF)	1	EBU	6372

Reaction No. 64103							
$\text{F-1} + \text{PaO2+1} = \text{PaO2+1_F-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.66(3SF)	1	EBU	6372

Reaction No. 64104							
$2\text{<F-1>} + \text{PaO2+1} = \text{PaO2+1_F-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.28(3SF)	1	EBU	6372

Reaction No. 64105							
$3\text{<F-1>} + \text{PaO2+1} = \text{PaO2+1_F-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.99(3SF)	1	EBU	6372

Reaction No. 64106							
$4\text{<F-1>} + \text{PaO2+1} = \text{PaO2+1_F-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.83(2SF)	1	EBU	6372

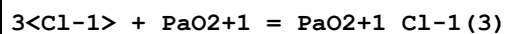
Reaction No. 64107							
$5\text{<F-1>} + \text{PaO2+1} = \text{PaO2+1_F-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.16(3SF)	1	EBU	6372

Reaction No. 64108							
$\text{Cl-1} + \text{PaO2+1} = \text{PaO2+1_Cl-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.16(2SF)	1	EBU	6372

Reaction No. 64109							
$2\text{<Cl-1>} + \text{PaO2+1} = \text{PaO2+1_Cl-1(2)}$							

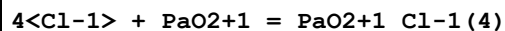
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.02 (1SF)	1	EBU	6372

Reaction No. 64110



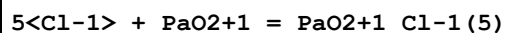
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.41 (2SF)	1	EBU	6372

Reaction No. 64111



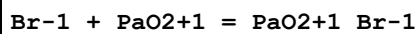
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.97 (2SF)	1	EBU	6372

Reaction No. 64112



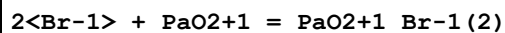
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.67 (3SF)	1	EBU	6372

Reaction No. 64113



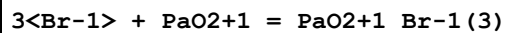
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.13 (2SF)	1	EBU	6372

Reaction No. 64114



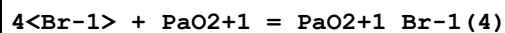
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.04 (1SF)	1	EBU	6372

Reaction No. 64115



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.40 (2SF)	1	EBU	6372

Reaction No. 64116



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.89 (2SF)	1	EBU	6372

Reaction No. 64117							
$5\text{<Br-1>} + \text{PaO2+1} = \text{PaO2+1_Br-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.49(3SF)	1	EBU	6372

Reaction No. 64118							
$\text{I-1} + \text{PaO2+1} = \text{PaO2+1_I-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.12(2SF)	1	EBU	6372

Reaction No. 64119							
$2\text{<I-1>} + \text{PaO2+1} = \text{PaO2+1_I-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.06(1SF)	1	EBU	6372

Reaction No. 64120							
$3\text{<I-1>} + \text{PaO2+1} = \text{PaO2+1_I-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.42(2SF)	1	EBU	6372

Reaction No. 64121							
$4\text{<I-1>} + \text{PaO2+1} = \text{PaO2+1_I-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.90(2SF)	1	EBU	6372

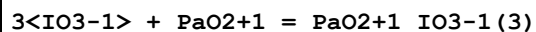
Reaction No. 64122							
$5\text{<I-1>} + \text{PaO2+1} = \text{PaO2+1_I-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.48(3SF)	1	EBU	6372

Reaction No. 64123							
$\text{IO3-1} + \text{PaO2+1} = \text{PaO2+1_IO3-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.37(2SF)	1	EBU	6372

Reaction No. 64124							
$2\text{<IO3-1>} + \text{PaO2+1} = \text{PaO2+1_IO3-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

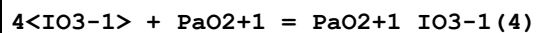
1	25	0	Inf. Dilution	lgK:-0.01(1SF)	1	EBU	6372
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Reaction No. 64125



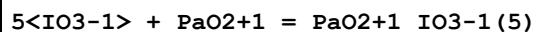
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.90(2SF)	1	EBU	6372

Reaction No. 64126



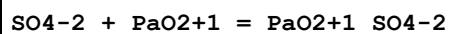
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.18(3SF)	1	EBU	6372

Reaction No. 64127



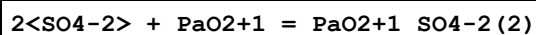
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.81(3SF)	1	EBU	6372

Reaction No. 64128



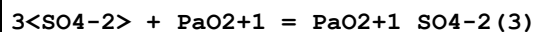
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.45(3SF)	1	EBU	6372

Reaction No. 64129



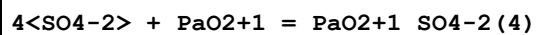
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.33(3SF)	1	EBU	6372

Reaction No. 64130



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.13(2SF)	1	EBU	6372

Reaction No. 64131



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.81(3SF)	1	EBU	6372

Reaction No. 64132

NH3 + PaO2+1 = PaO2+1_NH3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.55 (2SF)	1	EBU	6372

Reaction No. 64133							
2<NH3> + PaO2+1 = PaO2+1_NH3 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.44 (2SF)	1	EBU	6372

Reaction No. 64134							
3<NH3> + PaO2+1 = PaO2+1_NH3 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.20 (2SF)	1	EBU	6372

Reaction No. 64135							
4<NH3> + PaO2+1 = PaO2+1_NH3 (4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.32 (3SF)	1	EBU	6372

Reaction No. 64136							
5<NH3> + PaO2+1 = PaO2+1_NH3 (5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.89 (3SF)	1	EBU	6372

Reaction No. 64137							
NO3-1 + PaO2+1 = PaO2+1_NO3-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.21 (2SF)	1	EBU	6372

Reaction No. 64138							
2<NO3-1> + PaO2+1 = PaO2+1_NO3-1 (2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.50 (2SF)	1	EBU	6372

Reaction No. 64139							
3<NO3-1> + PaO2+1 = PaO2+1_NO3-1 (3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.86 (3SF)	1	EBU	6372

Reaction No. 64140							
$4\text{<NO3-1>} + \text{PaO2+1} = \text{PaO2+1_NO3-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.77(3SF)	1	EBU	6372

Reaction No. 64141							
$5\text{<NO3-1>} + \text{PaO2+1} = \text{PaO2+1_NO3-1(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-6.18(3SF)	1	EBU	6372

Reaction No. 64142							
$\text{PaO2+1} + \text{H+1_PO4-3} = \text{H+1_PaO2+1_PO4-3}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.37(3SF)	1	EBU	6372

Reaction No. 64143							
$\text{PaO2+1} + 2\text{<H+1_PO4-3>} = \text{H+1(2)_PaO2+1_PO4-3(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.26(3SF)	1	EBU	6372

Reaction No. 64144							
$\text{PaO2+1} + 3\text{<H+1_PO4-3>} = \text{H+1(3)_PaO2+1_PO4-3(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.92(3SF)	1	EBU	6372

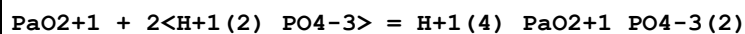
Reaction No. 64145							
$\text{PaO2+1} + 4\text{<H+1_PO4-3>} = \text{H+1(4)_PaO2+1_PO4-3(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.46(3SF)	1	EBU	6372

Reaction No. 64146							
$\text{PaO2+1} + 5\text{<H+1_PO4-3>} = \text{H+1(5)_PaO2+1_PO4-3(5)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.93(3SF)	1	EBU	6372

Reaction No. 64147							
$\text{PaO2+1} + \text{H+1(2)_PO4-3} = \text{H+1(2)_PaO2+1_PO4-3}$							

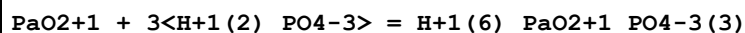
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.64 (2SF)	1	EBU	6372

Reaction No. 64148



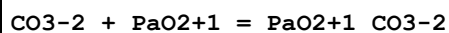
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.24 (2SF)	1	EBU	6372

Reaction No. 64149



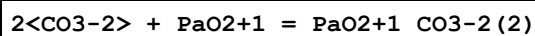
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.41 (3SF)	1	EBU	6372

Reaction No. 64150



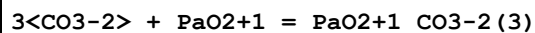
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.38 (3SF)	1	EBU	6372

Reaction No. 64151



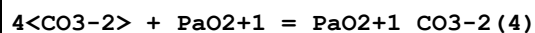
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.56 (3SF)	1	EBU	6372

Reaction No. 64152



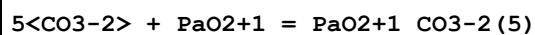
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.78 (3SF)	1	EBU	6372

Reaction No. 64153



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.14 (4SF)	1	EBU	6372

Reaction No. 64154



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.71 (4SF)	1	EBU	6372

Reaction No. 64155							
$\text{PaO2+1} + \text{H+1_CO3-2} = \text{H+1_PaO2+1_CO3-2}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.19(3SF)	1	EBU	6372

Reaction No. 64156							
$\text{PaO2+1} + 2<\text{H+1_CO3-2}> = \text{H+1(2)_PaO2+1_CO3-2(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.97(2SF)	1	EBU	6372

Reaction No. 64157							
$\text{PaO2+1} + 3<\text{H+1_CO3-2}> = \text{H+1(3)_PaO2+1_CO3-2(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.39(2SF)	1	EBU	6372

Reaction No. 64158							
$\text{PaO2+1} + 4<\text{H+1_CO3-2}> = \text{H+1(4)_PaO2+1_CO3-2(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.80(3SF)	1	EBU	6372

Reaction No. 64159							
$\text{SCN-1} + \text{PaO2+1} = \text{PaO2+1_SCN-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.26(2SF)	1	EBU	6372

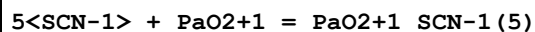
Reaction No. 64160							
$2<\text{SCN-1}> + \text{PaO2+1} = \text{PaO2+1_SCN-1(2)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.06(1SF)	1	EBU	6372

Reaction No. 64161							
$3<\text{SCN-1}> + \text{PaO2+1} = \text{PaO2+1_SCN-1(3)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.84(2SF)	1	EBU	6372

Reaction No. 64162							
$4<\text{SCN-1}> + \text{PaO2+1} = \text{PaO2+1_SCN-1(4)}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

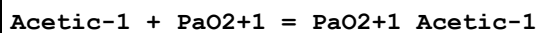
1	25	0	Inf. Dilution	lgK:-2.03 (3SF)	1	EBU	6372
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Reaction No. 64163



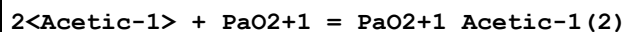
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.60 (3SF)	1	EBU	6372

Reaction No. 64164



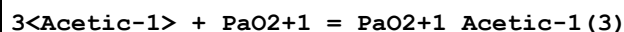
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.89 (2SF)	1	EBU	6372

Reaction No. 64165



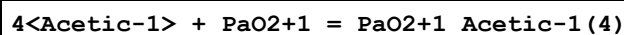
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.47 (2SF)	1	EBU	6372

Reaction No. 64166



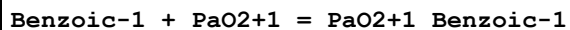
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.03 (3SF)	1	EBU	6372

Reaction No. 64167



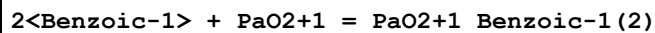
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.50 (3SF)	1	EBU	6372

Reaction No. 64168



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.08 (3SF)	1	EBU	6372

Reaction No. 64169



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.00 (3SF)	1	EBU	6372

Reaction No. 64170

3<Benzoic-1> + PaO2+1 = PaO2+1_Benzoic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.01(1SF)	1	EBU	6372

Reaction No. 64171							
4<Benzoic-1> + PaO2+1 = PaO2+1_Benzoic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.77(3SF)	1	EBU	6372

Reaction No. 64172							
5<Benzoic-1> + PaO2+1 = PaO2+1_Benzoic-1(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.31(3SF)	1	EBU	6372

Reaction No. 64173							
Cat-2 + PaO2+1 = PaO2+1_Cat-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.71(3SF)	1	EBU	6372

Reaction No. 64174							
2<Cat-2> + PaO2+1 = PaO2+1_Cat-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.88(4SF)	1	EBU	6372

Reaction No. 64175							
EtDiAm + PaO2+1 = PaO2+1_EtDiAm							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.99(3SF)	1	EBU	6372

Reaction No. 64176							
2<EtDiAm> + PaO2+1 = PaO2+1_EtDiAm(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.67(3SF)	1	EBU	6372

Reaction No. 64177							
Gly-1 + PaO2+1 = PaO2+1_Gly-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.54(3SF)	1	EBU	6372

Reaction No. 64178							
2<Gly-1> + PaO2+1 = PaO2+1_Gly-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.46(3SF)	1	EBU	6372

Reaction No. 64179							
Glycolic-1 + PaO2+1 = PaO2+1_Glycolic-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.75(2SF)	1	EBU	6372

Reaction No. 64180							
2<Glycolic-1> + PaO2+1 = PaO2+1_Glycolic-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.11(2SF)	1	EBU	6372

Reaction No. 64181							
3<Glycolic-1> + PaO2+1 = PaO2+1_Glycolic-1(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.68(3SF)	1	EBU	6372

Reaction No. 64182							
4<Glycolic-1> + PaO2+1 = PaO2+1_Glycolic-1(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.50(3SF)	1	EBU	6372

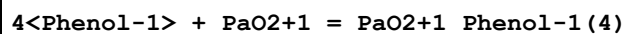
Reaction No. 64183							
Phenol-1 + PaO2+1 = PaO2+1_Phenol-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.00(3SF)	1	EBU	6372

Reaction No. 64184							
2<Phenol-1> + PaO2+1 = PaO2+1_Phenol-1(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.05(3SF)	1	EBU	6372

Reaction No. 64185							
3<Phenol-1> + PaO2+1 = PaO2+1_Phenol-1(3)							

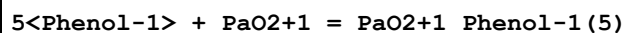
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.27(3SF)	1	EBU	6372

Reaction No. 64186



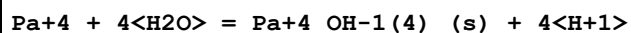
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.70(3SF)	1	EBU	6372

Reaction No. 64187



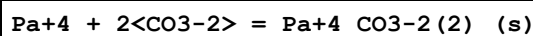
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.38(3SF)	1	EBU	6372

Reaction No. 65522



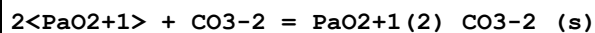
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.83(3SF)	4	EBU	6372

Reaction No. 65523



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.17(4SF)	4	EBU	6372

Reaction No. 65524



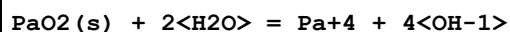
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.55(4SF)	4	EBU	6372

Reaction No. 66864



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.(2SF)	4	ENS	773
2	25	1	Unknown	lgK:-15.8(3SF)	4	ENS	773

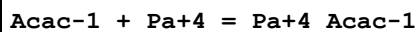
Reaction No. 66865



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-55.4(3SF)	1	ELC	

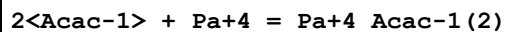
2	25	0	Inf. Dilution	lgK:-53.4 (3SF)	0	ENS	773
3	25	1	Unknown	lgK:-53.3 (3SF)	0	ENS	773

Reaction No. 70075



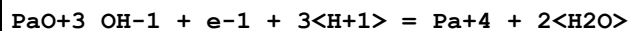
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene NaClO4	lgK:6.1 (2SF)	5	MGL	2856

Reaction No. 70076



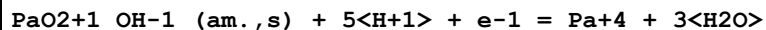
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Benzene NaClO4	lgK:13.2 (0.1SD)	5	MGL	2856

Reaction No. 72157



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-0.1 (1SF)	3	CRV	7761
2	25	0	Inf. Dilution	E:-0.1 (1SF)	4	CRV	22551
3	25	0	Inf. Dilution	dH:-85.3 (3SF) kJ	3	CRV	7761

Reaction No. 72158



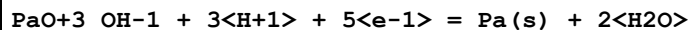
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.5 (2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.1 (1SF)	0	CRV	7761

Reaction No. 72159



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-9.6 (2SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.2 (1SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:15.6 (3SF) kJ	2	CRV	7761

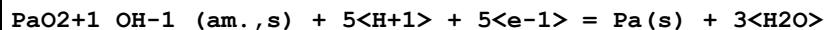
Reaction No. 72160



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
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1	25	0	Inf. Dilution	lgK:-102.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.21(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:550.7(4SF)kJ	2	CRV	7761

Reaction No. 72161



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-103.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.22(3SF)	0	CRV	7761

Reaction No. 72162



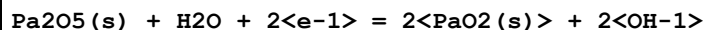
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.24(3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:1090.(4SF)kJ	3	CRV	7761

Reaction No. 72163



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-99.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.50(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:536.3(4SF)kJ	2	CRV	7761

Reaction No. 72164



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-37.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.0(2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:123.9(4SF)kJ	2	CRV	7761

Reaction No. 72165



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.2(2SF)	0	CRV	7761

Reaction No. 72166

PaO2+1_OH-1_(am.,s) + e-1 = PaO2(am.,s) + OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.3(2SF)	0	CRV	7761

Reaction No. 72167							
PaO+3_OH-1(4) + H2O + 5<e-1> = Pa(s) + 6<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.04(3SF)	3	CRV	7761

Reaction No. 72168							
PaO2+1_OH-1_(am.,s) + 2<H2O> + 5<e-1> = Pa(s) + 5<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.05(3SF)	3	CRV	7761

Reaction No. 72169							
Pa2O5(s) + 5<H2O> + 10<e-1> = 2<Pa(s)> + 10<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-349.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.07(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:1649.2(5SF)kJ	2	CRV	7761

Reaction No. 72170							
PaO2(am.,s) + 2<H2O> + 4<e-1> = Pa(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-2.24(3SF)	3	CRV	7761
2	25	0	Inf. Dilution	dH:760.(3SF)kJ	3	EES	
Note (2): By analogy with the crystalline solid							

Reaction No. 72171							
PaO2(s) + 2<H2O> + 4<e-1> = Pa(s) + 4<OH-1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-155.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-2.33(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:760.0(4SF)kJ	2	CRV	7761

Reaction No. 75595							
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2<Br2(l)> + Pa(s) = Pa+4_Br-1(4)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-787.8(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-824.0(4SF)kJ	3	CRV	6330

Reaction No. 75625							
2<Cl2(g)> + Pa(s) = Pa+4_Cl-1(4)_ (s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:168.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-953.0(4SF)kJ	0	CRV	6330
3	25	0	Inf. Dilution	dH:-1043.0(5SF)kJ	2	CRV	6330

Reaction No. 75943							
Pa+4_Oxalic-2 + Oxalic-2 = Pa+4_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.60(3SF)	5	EOD	11516
Note (1): Moskvina A et al., Zhur. Neorg. Khim., 1967, 12, 359							

Reaction No. 75944							
Pa+5_Oxalic-2 + Oxalic-2 = Pa+5_Oxalic-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	1	KNO3	lgK:0.70(2SF)	5	MIX	4421
2	25	0	Inf. Dilution	lgK:0.7(1SF)	1	ESC	

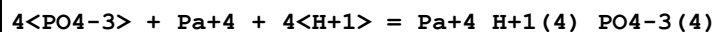
Reaction No. 78808							
PO4-3 + Pa+4 + H+1 = Pa+4_H+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.97(4SF)	1	ELC	

Reaction No. 78809							
2<PO4-3> + Pa+4 + 2<H+1> = Pa+4_H+1(2)_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:48.33(4SF)	1	ELC	

Reaction No. 78810							
3<PO4-3> + Pa+4 + 3<H+1> = Pa+4_H+1(3)_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

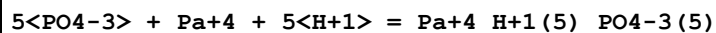
1	25	0	Inf. Dilution	lgK:70.36(4SF)	1	ELC	
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Reaction No. 78811



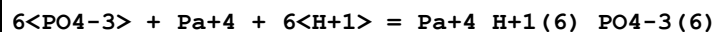
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:91.13(4SF)	1	ELC	

Reaction No. 78812



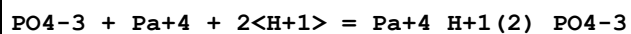
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:110.7(4SF)	1	ELC	

Reaction No. 78813



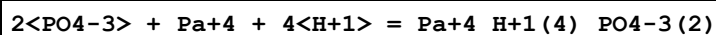
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:129.1(4SF)	1	ELC	

Reaction No. 78814



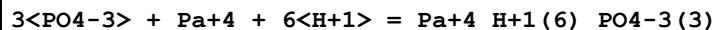
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.17(4SF)	1	ELC	

Reaction No. 78815



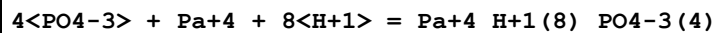
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:46.75(4SF)	1	ELC	

Reaction No. 78816



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:67.97(4SF)	1	ELC	

Reaction No. 78817



No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:87.95(4SF)	1	ELC	

Reaction No. 78818

5<PO4-3> + Pa+4 + 10<H+1> = Pa+4_H+1(10)_PO4-3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:106.7(4SF)	1	ELC	

Reaction No. 78819							
6<PO4-3> + Pa+4 + 12<H+1> = Pa+4_H+1(12)_PO4-3(6)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:124.4(4SF)	1	ELC	

Reaction No. 78820							
PO4-3 + PaO2+1 + H+1 = H+1_PaO2+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.7(4SF)	1	ELC	

Reaction No. 78821							
2<PO4-3> + PaO2+1 + 2<H+1> = H+1(2)_PaO2+1_PO4-3(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:29.91(4SF)	1	ELC	

Reaction No. 78822							
3<PO4-3> + PaO2+1 + 3<H+1> = H+1(3)_PaO2+1_PO4-3(3)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.9(4SF)	1	ELC	

Reaction No. 78823							
4<PO4-3> + PaO2+1 + 4<H+1> = H+1(4)_PaO2+1_PO4-3(4)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:54.77(4SF)	1	ELC	

Reaction No. 78824							
5<PO4-3> + PaO2+1 + 5<H+1> = H+1(5)_PaO2+1_PO4-3(5)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:65.56(4SF)	1	ELC	

Reaction No. 78825							
PO4-3 + PaO2+1 + 2<H+1> = H+1(2)_PaO2+1_PO4-3							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.16(4SF)	1	ELC	

Reaction No. 78826							
$2\text{<PO4-3>} + \text{PaO2+1} + 4\text{<H+1>} = \text{H+1(4)}_{\text{PaO2+1_PO4-3(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.8(4SF)	1	ELC	

Reaction No. 78827							
$3\text{<PO4-3>} + \text{PaO2+1} + 6\text{<H+1>} = \text{H+1(6)}_{\text{PaO2+1_PO4-3(3)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.15(4SF)	1	ELC	

Reaction No. 80369							
$2\text{<NTA-3>} + \text{PaO+3} = \text{PaO+3}_{\text{NTA-3(2)}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	(H,Na)ClO4	lgK:28.6(3SF)	2	MDS	30786
Note (1): lgK varies with acidity							

Reaction No. 80589							
$\text{PaO+3}_{\text{OH-1}} + \text{H2O} = \text{PaO+3}_{\text{OH-1(2)}} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.260(0.075SD)	5	CRV	31301

Reaction No. 80590							
$\text{PaO+3}_{\text{OH-1(2)}} + 2\text{<H2O>} = \text{Pa+5}_{\text{OH-1(5)}} + \text{H+1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.150(0.2SD)	5	CRV	31301

Reaction No. 80591							
$\text{PaO+3}_{\text{OH-1}} + \text{Cl-1} = \text{PaO+3}_{\text{Cl-1}_{\text{OH-1}}}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.567(0.01SD)	5	CRV	31301

Reaction No. 80592							
$\text{PaO+3}_{\text{OH-1}} + \text{SO4-2} + \text{H+1} = \text{PaO+3}_{\text{SO4-2}} + \text{H2O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.890(0.09SD)	5	CRV	31301

Reaction No. 80593							
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PaO+3_OH-1 + 2<SO4-2> + H+1 = PaO+3_SO4-2(2) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.000(0.1SD)	5	CRV	31301

Reaction No. 80594							
PaO+3_OH-1 + 3<SO4-2> + H+1 = PaO+3_SO4-2(3) + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.590(0.115SD)	5	CRV	31301

Reaction No. 81102							
O2(g) + Pa(s) - e-1 = PaO2+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:185.8(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.22(1.70SD)	0	CRV	32222

Reaction No. 81103							
PaO2+1 + H+1 = PaO+3_OH-1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.50(0.10SD)	4	CRV	32222

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EBU	Brown's Unified Theory
EES	Personal estimate
ELC	Linear Combination of Reactions
ENS	Not specified
EOD	Calculated from data of others
EOR	Other Reaction Constants
ESC	Standard conditions anchor
MCE	Cation exchange
MDG	Combination of Thermodynamic Data
MDS	Distribution between two phases
MGL	Glass electrode
MIX	Ion exchange
MQH	Quinhydrone electrode

References

Reference No.: 437(80BoH)

IUPAC Chemical Data Series No. 27, 1980 Bond AM; Hefter GT; Critical Survey of Stability Constants and Related Thermodynamic Data of Fluoride Complexes in Aqueous Solution; Pergamon, Oxford, U.K.

Reference No.: 552(95SMM)

NIST Critical Stability Constants of Metal Complexes Database, 1995 Smith RM; Martell AE; Motekaitis RJ; Version 2.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 773(85KoS)

Handbook of Chemical Equilibria in Analytical Chemistry, 1985 Kotrly S; Sucha L; Ellis Horwood Series in Anal. Chem.; Eds. Chalmers RA & Masson M; Ellis Horwood, Chichester

Reference No.: 775(71CCM)

Selected Constants, 1971 Charlot G; Collumeau A; Marchon MJC; Oxidation-Reduction Potentials of Inorganic Substances in Aqueous Solution; Butterworths, London

Reference No.: 906(79Per)

IUPAC Chemical Data Series No. 22, 1979 Perrin DD; Stability Constants of Metal-Ion Complexes, First Edition Part B, Organic Ligands; Pergamon, Oxford

Reference No.: 931(71SiM)

Stability Constants of Metal-ion Complexes SP25, 1971 Sillen LG; Martell AE; Supplement No 1 to Special Publication No 25; Chemical Society, London

Reference No.: 2556(93MSM)

NIST Critical Stability Constants of Metal Complexes Database, 1993 Martell AE; Smith RM; Motekaitis RJ; Version 1.0; National Institute of Standards and Technology, Gaithersburg, USA

Reference No.: 2856(74Lun)

Acta Chem. Scand., 1974, A28, 243 - 247; Lundqvist R; Aqueous Chemistry of Protactinium(IV). 1. Stability Constants for Pa(IV) - Acetylacetone Complexes

Reference No.: 3006(85BPJ)

Standard Potentials in Aqueous Solution, 1985 Bard AJ; Parsons R; Jordan J; Monographs in Electroanalytical Chemistry and Electrochemistry; Marcel Dekker Inc., New York and Basel

Reference No.: 4421(66DMP)

Russ. J. Inorg. Chem., 1966, 11, 702 - 706; Davydov AV; Marov IN; Palei PN; Complex Formation by Protactinium in Oxalate, Phosphate, and Arsenate Solutions

Reference No.: 5156(91ReB)

Radiochim. Acta, 1991, 52/53, 453 - 456; Read D; Broyd TW; Recent Progress in Testing Chemical Equilibrium Models: The CHEMVAL Project

Reference No.: 5256(91MiK)

Coord. Chem. Rev., 1991, 109, 1 - 5; Mikhchev NB; Kamenskaya AN; Complex Formation of the Lanthanides and Actinides in Lower Oxidation States

Reference No.: 5398(82Hog)

IUPAC Chemical Data Series No. 21, 1982 Hogfeldt E; Stability Constants of Metal-ion Complexes, 2nd Suppl., Part A, Inorganic Ligands; Pergamon, Oxford

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 6372(87BrW)

OECD NEA Report, 1987 Brown PL; Wanner H; Predicted Formation Constants Using the Unified Theory of Metal Ion Complexation; Nuclear Energy Authority, Paris

Reference No.: 6478(81BaM)

Am. J. Sci., 1981, 281, 935 - 962; Baes CF Jr; Mesmer RE; The Thermodynamics of Cation Hydrolysis

Reference No.: 7276(88CoW)

Advanced Inorganic Chemistry, 5th Edn., 1988 Cotton FA; Wilkinson G; John Wiley & Sons Inc., N.Y., U.S.A

Reference No.: 7761(89Bra)

J. Phys. Chem. Ref. Data, 1989, 18, 1 - 21; Bratsch SG; Standard Electrode Potentials and Temperature Coefficients in Water at 298.15 K

Reference No.: 11516(00PeP)

IUPAC Stability Constants Database, 2000 Pettit LD; Powell HKJ; Release 4; Academic Software, U.K.

Reference No.: 22551(06Inz)

Encyclopedia of Electrochemistry, Vol. 7a, 2006, 17 - 75; Inzelt G; Standard, Formal and Other Characteristic Potentials of Selected Electrode Reactions

Reference No.: 30786(18LLA)

New J. Chem., 2018, 42, 7789 - 7795; Luchini C; Leguay S; Aupiais J; Cannes C; Le Naour C; Complexation of protactinium(V) with nitrilotriacetic acid: a study at the tracer scale

Reference No.: 31301(10KFD)

Japan Atomic Energy Agency Data/Code 2009-024, 2010 Kitamura A; Fujiwara K; Doi R; Yoshida Y; Mihara M; Terashima M; Yui M; JAEA Thermodynamic Database for Performance Assessment of Geological Disposal of High-level Radioactive and TRU Wastes; Geological Isolation Research Unit, JAEA, Japan

Reference No.: 32222(06DGC)

Technical Report TR-O6-17, 2006 Duro L; Grive M; Cera E; Domenech C; Bruno J; Update of a Thermodynamic Database for Radionuclides to Assist Solubility Limits Calculation for Performance Assessment; Swedish Nucl. Fuel & Waste Management Co, Stockholm, Sweden